Yuxin Jin

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EDUCATION

Northwest University, Xi'an, China **09/2024 - Present**

MSc in Computer Science and Technology

- GPA: 3.3/4.0
- Relevant Coursework: Multiscale Image Analysis, Graph Neural Networks, Multimodal Image Analysis.

Xi'an University of Technology, Xi'an, China **09/2020 - 06/2024**

BSc in Artificial Intelligence

- GPA: 3.57/5.0
- Relevant Coursework: Discrete Mathematics, Functional Analysis, Algorithm Analysis and Design, Fundamentals of Machine Learning, Computational Intelligence, Neural Networks and Deep Learning.

RESEARCH INTERESTS

- Causal Machine Learning, Causal Representation Learning, Causal Discovery, Multi-Environment Causal Learning, Learning under Distribution Shifts, and AI for Medical Imaging.

PUBLICATIONS & MANUSCRIPTS

Jin, Y., Zhao, F., Chen, Y., & He, X. (2026). Uncertainty-Guided Curriculum Learning for Automated Liver Fibrosis Staging on Heterogeneous MRI. In Comprehensive Analysis and Computing of Real-World Medical Images: CARE 2025, Lecture Notes in Computer Science (LNCS), Vol. 16257, Springer Nature.

- **First author and oral presenter** at the CARE 2025 Workshop (MICCAI 2025); received the **Best Paper Runner-up Award** and ranked in the **top 16%** of submissions.
- Proposed an **uncertainty-guided curriculum learning framework** for liver fibrosis staging on heterogeneous multi-center and multi-sequence MRI under distribution shifts.
- Designed a sample-level uncertainty measure to construct an easy-to-hard training curriculum, improving robustness to difficult cases and cross-sequence distribution shifts.

Liu, Y., Shi, N., Zhang, Z., Zhang, H., Wang, B., Yu, D., Wang, N., **Jin, Y**., et al. (2026). How Far Has AI Come in Liver Fibrosis Staging? A Large-Scale Real-World Dataset and Benchmark. **Submitted to Medical Image Analysis.**

- Contributed the uncertainty-guided curriculum learning method above as one of **nine independently developed AI approaches** evaluated in the large-scale LiFS benchmark.
- Contributed to the evaluation of liver fibrosis staging under **real-world multi-center, multi-scanner, and multi-sequence MRI heterogeneity**, including in-distribution and cross-center generalization.

Zheng, Z., **Jin, Y**., Gao, H., He, X., & He, X. (2026). AOAsNet: A model exploring the correlation of the attention modules for medical image classification. In Proceedings of the 48th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC 2026), Toronto, Canada.

- Developed an attention-correlation framework that models the relationship between attention-enhanced features and original feature representations for medical image classification.
- Designed a probability distribution-based correlation module and validated the framework on three public medical imaging datasets with strong accuracy and computational efficiency.

RESEARCH EXPERIENCE

Causal Representation Learning under Structured Process-Informed Environment Shifts **03/2026 - Present**
Current Research Project

- Investigating **multi-environment causal representation learning** where structured process information is used to characterize environment-dependent mechanism shifts.
- Developing latent representation and causal discovery methods to separate **process-driven variation from disease-related causal structure in high-dimensional imaging data**.
- Studying whether structured environment information can improve **causal identifiability and robustness under distribution shifts**.

Robust Multimodal Classification of OLPD with Missing Modalities **12/2024 - 05/2025**
Research Project

- Formulated oral potentially malignant lesion (OLPD) diagnosis as a **multimodal classification problem with missing modalities** and developed a robust medical image classification framework.
- Designed a deep learning pipeline integrating **missing-modality modeling, multimodal feature fusion, and uncertainty-aware classification**, and implemented the complete training and evaluation process.
- Conducted quantitative and qualitative evaluations to verify robustness under modality-missing scenarios; the project received the **Second Prize in a university-level academic competition**.

Pseudo-Siamese Transformer–CNN Framework for Medical Image Generation **04/2025 - 10/2025**
Research Project

- Proposed **SiamTC-GAN**, a pseudo-Siamese generative framework combining Transformer and CNN modules to recover missing medical imaging modalities while capturing both global structure and local details.
- Designed a **multi-head cross-modal attention mechanism (MHTA)** and a hybrid enhancement loss (HE-loss) to improve visual quality and cross-modal semantic consistency.
- Evaluated the model on contrast-enhanced CT and MRI datasets, achieving improved **FID, MAE, PSNR, and SSIM** compared with baseline methods.

Evolutionary Optimization for Constrained Multimodal Multi-Objective Problems **02/2024 - 06/2024**
Undergraduate Thesis

- Proposed an improved **differential evolution algorithm** with a species differentiation mechanism for constrained multimodal multi-objective optimization.
- Designed adaptive clustering, feasibility balancing, and a dual-stage convergence–diversity strategy to maintain multiple equivalent Pareto solution sets under complex constraints.
- Evaluated the algorithm on **17 benchmark functions**, demonstrating improved convergence and solution diversity compared with baseline methods.

AWARDS & QUALIFICATIONS

- Excellent Award, "Challenge Cup" College Student Entrepreneurship Plan Competition, School of Electronic Information, Northwest University **2025**
- The Best Paper Runner-Up Award, CARE 2025 Official Challenge, 28th International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI) **2025**
- Second Prize (University Level), Graduate Electronic Design Contest (Technical Track), Northwest University **2025**
- Second Prize (University Level), 12th "Challenge Cup" Extracurricular Academic Science and Technology Works Competition **2025**
- "Shangzhen Duxue" Science and Technology Advanced Individual Honor, Xi'an University of Technology **2024**
- Shangneng Xiumei" Advanced Individual Honor, Xi'an University of Technology **2023**
- First Prize (Provincial Level), College Student Innovation and Entrepreneurship Training Program **2023**

SKILLS & LANGUAGES

- **Programming:** Python, MATLAB, C++
- **Machine Learning:** PyTorch, scikit-learn
- **Scientific Computing:** NumPy, pandas
- **Tools:** Git, LaTeX
- **Languages:** Mandarin (Native), English (Proficient)